

файл test.c

```
#include <linux/module.h>
#include <linux/kernel.h>
#include <linux/fs.h>
#include <linux/rwlock.h>
#include <linux/proc_fs.h>
#include <linux/sysfs.h>
#include <linux/string.h>
#include <linux/kobject.h>

static int major = 0;
static struct proc_dir_entry *test = NULL;
static struct kobject *test_kobj = NULL;
static rwlock_t lock;
static char test_string[15] = "Hello\n\0";

static ssize_t test_read(struct file *fd, char __user *buff, size_t size, loff_t *off)
{
    size_t rc = 0;

    read_lock(&lock);
    rc = simple_read_from_buffer(buff, size, off, test_string, 15);
    read_unlock(&lock);

    return rc;
}

static ssize_t test_write(struct file *fd, const char __user *buff, size_t size, loff_t
*off)
{
    size_t rc = 0;

    if (size > 15)
        return -EINVAL;

    write_lock(&lock);
    rc = simple_write_to_buffer(test_string, 15, off, buff, size);
    write_unlock(&lock);

    return rc;
}

static ssize_t test_proc_read(struct file *fd, char __user *buff, size_t size, loff_t
*off)
```

```

{
    size_t rc = 0;

    rc = simple_read_from_buffer(buff, size, off, test_string, 15);

    return rc;
}

static ssize_t test_proc_write(struct file *fd, const char __user *buff, size_t size,
loff_t *off)
{
    size_t rc = 0;

    if (size > 15) return -EINVAL;

    rc = simple_write_to_buffer(test_string, 15, off, buff, size);

    return rc;
}

static ssize_t string_show(struct kobject *kobj, struct kobj_attribute *attr,
                           char *buf)
{
    size_t rc = 0;
    memcpy(buf, test_string, 15);
    rc = strlen(test_string);

    return rc;
}

static ssize_t string_store(struct kobject *kobj, struct kobj_attribute *attr,
                           char const *buf, size_t count)
{
    size_t rc = 0;

    if (count >= 15)
        return -EINVAL;

    memcpy(test_string, buf, 15);
    rc = strlen(test_string);

    return rc;
}

static struct file_operations fops = {

```

```

.owner = THIS_MODULE,
.read = test_read,
.write = test_write};

// for kernel ver >= 5.x.x
static const struct proc_ops pops = {
    .proc_read = test_proc_read,
    .proc_write = test_proc_write,
};

// for kernel ver < 5.x.x
// static const struct file_operations pops = {
//     .read = test_proc_read,
//     .write = test_proc_write
// };

static struct kobj_attribute string_attribute =
    __ATTR(test_string, 0644, string_show, string_store);

static struct attribute *attrs[] = {
    &string_attribute.attr,
    NULL,
};

static struct attribute_group attr_group = {
    .attrs = attrs,
};

int init_module(void)
{
    int retval = 0;
    pr_info("My test module loaded!!!\n");
    rwlock_init(&lock);
    major = register_chrdev(major, "test", &fops);

    if (major < 0)
        return major;

    pr_info("Major = %d\n", major);

    test = proc_create("test", 0666, NULL, &pops);

    test_kobj = kobject_create_and_add("kobject_test", kernel_kobj);

    if (!test_kobj)

```

```

        return -ENOMEM;

    retval = sysfs_create_group(test_kobj, &attr_group);

    if (retval)
    {
        kobject_put(test_kobj);
        return retval;
    }

    return 0;
}

void cleanup_module(void)
{
    unregister_chrdev(major, "test");
    proc_remove(test);
    kobject_put(test_kobj);
    pr_info("My test module unloaded!!!\n");
}

MODULE_LICENSE("GPL");

```

Сборка файла test.c

Make

```

• tair@tair-Bravo-15-C7VE:~/Eltex/lesson20/file_device_proc_sys$ make
make -C /lib/modules/6.17.0-35-generic/build M=/home/tair/Eltex/lesson20/file_device_proc_sys modules
make[1]: вход в каталог «/usr/src/linux-headers-6.17.0-35-generic»
make[2]: вход в каталог «/home/tair/Eltex/lesson20/file_device_proc_sys»
warning: the compiler differs from the one used to build the kernel
  The kernel was built by: x86_64-linux-gnu-gcc-13 (Ubuntu 13.3.0-6ubuntu2~24.04.1) 13.3.0
  You are using:          gcc-13 (Ubuntu 13.3.0-6ubuntu2~24.04.1) 13.3.0
  CC [M]  test.o
  MODPOST Module.symvers
WARNING: modpost: missing MODULE_DESCRIPTION() in test.o
  CC [M]  test.mod.o
  CC [M]  .module-common.o
  LD [M]  test.ko
  BTF [M] test.ko
Skipping BTF generation for test.ko due to unavailability of vmlinux
make[2]: выход из каталога «/home/tair/Eltex/lesson20/file_device_proc_sys»
make[1]: выход из каталога «/usr/src/linux-headers-6.17.0-35-generic»

```

Загрузка модуля

```
tair@tair-Bravo-15-C7VE:~/Eltex/lesson20/file_device_proc_sys$ sudo insmod test.ko
[sudo] пароль для tair:
tair@tair-Bravo-15-C7VE:~/Eltex/lesson20/file_device_proc_sys$ sudo dmesg | tail -n10
[ 8168.290838] No Arguments are initialized for method [_Q9A]

[ 8168.290841] ACPI Error: Aborting method \_SB.PCI0.SBRG.EC._Q9A due to previous error (AE_NO_HANDLER)
[ 8171.547593] [UFW BLOCK] IN=enol OUT= MAC=01:00:5e:00:00:fb:c0:06:c3:97:eb:31:08:00 SRC=192.168.1.1
[ 8172.044556] [UFW BLOCK] IN=enol OUT= MAC=01:00:5e:00:00:fb:a4:2b:b0:eb:cf:e1:08:00 SRC=192.168.1.1
[ 8229.364968] audit: type=1400 audit(1781633268.625:279): apparmor="AUDIT" operation="usersns_create"
file="unconfined" pid=57243 comm="bwrap" requested="usersns_create" target="unprivileged_usersns"
[ 8229.365876] audit: type=1400 audit(1781633268.626:280): apparmor="DENIED" operation="capabilities_setpcap"
lity=8 capname="setpcap"
[ 8229.365945] audit: type=1400 audit(1781633268.626:281): apparmor="DENIED" operation="capabilities_setpcap"
lity=12 capname="net_admin"
[ 8255.619552] My test module loaded!!!
[ 8255.619560] Major = 504
tair@tair-Bravo-15-C7VE:~/Eltex/lesson20/file_device_proc_sys$
```

Файл test в папке proc

Левая панель	Файл	Команда	Настройки	Правая панель
< /proc .[^]>				
Имя			Размер	Время правки
.и				
mdstat			0	июн 16 22:50
meminfo			0	июн 16 22:50
misc			0	июн 16 22:50
modules			0	июн 16 22:50
@mounts			11	июн 16 22:50
mtrr			0	июн 16 22:50
pagetypeinfo			0	июн 16 22:50
partitions			0	июн 16 22:50
schedstat			0	июн 16 22:50
slabinfo			0	июн 16 22:50
softirqs			0	июн 16 22:50
stat			0	июн 16 22:50
swaps			0	июн 16 22:50
sysrq-trigger			0	июн 16 22:50
test			0	июн 17 01:09
timer_list			0	июн 16 22:50
uptime			0	июн 16 22:50
version			0	июн 16 22:50
version_signature			0	июн 16 22:50
vmallocinfo			0	июн 16 22:50
vmstat			0	июн 16 22:50
zoneinfo			0	июн 16 22:50
test				

Директория `kobject_test` и файл `test_string` файловой системы `sys`

The screenshot displays a file manager window with a blue header bar containing the following tabs: "Левая панель", "Файл", "Команда", "Настройки", and "Правая панель". The address bar shows the path `/sys/kernel/kobject_test`. Below the header, a table lists the contents of the directory:

Имя	Размер	Время правки
<code>..</code>	<code>-ВВЕРХ-</code>	июн 16 22:50
<code>test_string</code>	4096	июн 17 01:13

The status bar at the bottom of the window displays `-ВВЕРХ-`.

Тестирование

Proc:

```
● tair@tair-Bravo-15-C7VE:~/Eltex/lesson20/file_device_proc_sys$ cd /proc/
● tair@tair-Bravo-15-C7VE:/proc$ cat test
Hello
● tair@tair-Bravo-15-C7VE:/proc$ echo "Hi" > test
● tair@tair-Bravo-15-C7VE:/proc$ cat test
Hi
lo
○ tair@tair-Bravo-15-C7VE:/proc$
```

Sys:

```

• tair@tair-Bravo-15-C7VE:/sys/kernel/kobject_test$ cat test_string
Hi
lo
○ tair@tair-Bravo-15-C7VE:/sys/kernel/kobject_test$ echo "Hi" > test_string
bash: test_string: Отказано в доступе
○ tair@tair-Bravo-15-C7VE:/sys/kernel/kobject_test$

```

Дерегистрация модуля

```
● tair@tair-Bravo-15-C7VE:~/Eltex/lesson20/file_device_proc_sys$ sudo rmmod test
[sudo] пароль для tair:
● tair@tair-Bravo-15-C7VE:~/Eltex/lesson20/file_device_proc_sys$ dmesg | tail -n5
dmesg: чтение буфера ядра завершилось неудачно: Операция не позволена
● tair@tair-Bravo-15-C7VE:~/Eltex/lesson20/file_device_proc_sys$ sudo dmesg | tail -n5
[ 8767.544055] [UFW BLOCK] IN=en01 OUT= MAC=01:00:5e:00:00:01:c0:06:c3:97:eb:31:08:00
[ 8772.540690] [UFW BLOCK] IN=en01 OUT= MAC=01:00:5e:00:00:fb:a4:2b:b0:eb:cf:e1:08:00
[ 8796.543878] [UFW BLOCK] IN=en01 OUT= MAC=01:00:5e:00:00:fb:c0:06:c3:97:eb:31:08:00
[ 8797.040448] [UFW BLOCK] IN=en01 OUT= MAC=01:00:5e:00:00:fb:a4:2b:b0:eb:cf:e1:08:00
[ 8818.244073] My test module unloaded!!!
○ tair@tair-Bravo-15-C7VE:~/Eltex/lesson20/file_device_proc_sys$ █
```